

National Exam of Higher National Diploma-New program – 2021 Session

Spécialty/option : Software Engineering (SWE)
Paper : Computer Technology
Duration : 4Hours

Credits : 6

Instructions: Answer all questions

SECTION A: MCQ (40 MARKS).

1. Which of the following correctly declares an array ?
A) int array[10]; B) int array;
B) array{10}; D) array array[10];
2. What is the index number of the last element of an array with 9 elements?
A) 9 B) 8
C) 0 D) Programmer-defined
3. Which of the following accesses the seventh element stored in array?
A) array[6]; B) array[7];
C) array(7); D) array;
4. Where does the execution of the program starts?
A) user-defined function B) main function
C) void function D) none of the mentioned
5. What is meant by multiple inheritance?
A) deriving a base class from derived class
B) deriving a derived class from base class
C) deriving a derived class from more than one base class
D) none of the mentioned
6. What does polymorphism in OOPs mean?
A) Concept of allowing overriding of functions
B) Concept of hiding data
C) Concept of keeping things in different modules/files
D) Concept of wrapping things into a single unit
7. Which concept allows you to reuse the written code?
A) Encapsulation B) Abstraction
C) Inheritance D) Polymorphism

8. Which of the following explains Polymorphism?
- A)
`int func(int, int);`
`float func1(float, float);`
- B)
`int func(int);`
`int func(int);`
- C)
`int func(float);`
`float func(int, int, char);`
- D)
`int func();`
`int new_func();`
9. Which of the following shows multiple inheritances?
- A) $A \rightarrow B \rightarrow C$ B) $A \rightarrow B; A \rightarrow C$
C) $A, B \rightarrow CD$ B) $B \rightarrow A$
10. Which of the following is / are the Characteristics of information?
- A) Accuracy and Relevance B) Form of information and Timeliness
C) Completeness and Purpose D) All A, B & C
11. level supply information to strategic tier for the use of top management.
- A) Operational B) Environmental
C) Competitive D) Tactical
12. In the relational modes, cardinality is termed as:
- A) Number of tuples. B) Number of attributes.
C) Number of tables. D) Number of constraints.
13. Relational calculus is a
- A) Procedural language. B) Non- Procedural language.
C) Data definition language. D) High level language.
14. Cartesian product in relational algebra is
- A) a Unary operator. B) a Binary operator. C) a Ternary operator. D) not defined.
15. DML is provided for
- A) Description of logical structure of database.
B) Addition of new structures in the database system.
C) Manipulation & processing of database.
D) Definition of physical structure of database system.

16. 'AS' clause is used in SQL for
 A) Selection operation. B) Rename operation.
 C) Join operation. D) Projection operation.
17. ODBC stands for
 A) Object Database Connectivity. B) Oral Database Connectivity.
 C) Oracle Database Connectivity. D) Open Database Connectivity.
18. Architecture of the database can be viewed as
 A) two levels. B) four levels. C) three levels. D) one level.
19. In a relational model, relations are termed as
 A) Tuples. B) Attributes C) Tables. D) Rows.
20. Which of the following is an OSI layer one Device?
 A. Router
 B. Switch
 C. Hub
 D. Bridge
21. Which data unit exist in the OSI layer two?
 A. Packets
 B. Bits
 C. Frames
 D. All of the above
22. The set of data that the router consults before making decisions on the packets received is called?
 A. Decision table
 B. Decision support system
 C. Routing decision table
 D. Routing table
23. What is the maximum distance of a UTP cable?
 A. 10km
 B. 100m
 C. 100km
 D. None of the above
24. A program created with the intension to destroy the normal functioning of the computer is called?
 A. A virus
 B. A worm
 C. A malware
 D. A spyware

25. The unique address to a resource in a network such as the internet is called?
- A. Email address
 - B. IP address
 - C. MAC address
 - D. URL
26. This tool is always used to troubleshoot a network for problems
- A. Netstat
 - B. Ping
 - C. Ncpa.cpl
 - D. Ipconfig
27. It permits us to see the NIC adapter settings and all the configure settings in windows OS
- A. Ipconfig
 - B. Ipconfig /release
 - C. Ipconfig /all
 - D. Ifconfig
28. It is a communication media that is best used for very high bandwidth demands
- A. Coaxial cable
 - B. ADSL
 - C. XDSL
 - D. Optical Fiber
29. What protocol is used for the deployment of web services
- A. http
 - B. www
 - C. https
 - D. both A and C
 - E.
30. To refer to an external CSS file within an HTML document, we use:
- A. <CSS ...> ... </CSS>
 - B. <LINK ...> ... </LINK>
 - C. <Script...> ... </Script>
 - D. <LINK />
31. Communication between client and server is usually initiated by
- A. The client
 - B. http protocol
 - C. The server
 - D. Client and server
32. In hypertext, an anchor
- A. Is a link to another website
 - B. Links to a specific part of the same page
 - C. Links to another page in the same website
 - D. Links to the first link of a page
33. Which of the following is not true about tags in html
- A. Is a keyword embedded between < >
 - B. A tag can be open only or open close
 - C. Tags can be single line or multiline (block)
 - D. All open only tags are multiline

3. Describe four general strategies for dealing with deadlocks. (4marks)
4. Differentiate between a Kernel mode and a user mode of operating system. (2marks)

DatabaseSystem and Administration (20 marks)

A hospital management system contains a database to manage patient. For each patient, data about the symptoms that the patient shows is registered: fever, headache, cough, chest pains,.... Symptoms can have different severity: low, middle, or high. A patient may show several symptoms, e.g., high fever, medium headache and some cough. The database also contains data about diseases. Each disease is characterized by different symptoms: a patient with a cold should have fever and a cough, a malaria patient should have fever and fits of shivering, etc.

- 1) Describe this system in an E/R model. (7 marks)
- 2) Recall the rules to translate an E/R model into relational model (6 marks)
- 3) Translate the E/R model into a relational model. (7 marks)

Networking (10 marks)

1. What is the difference between packet-switched and circuit-switched networks? (2 marks)
2. Explain what a MAC addresses is (1 mark)
3. Define Single-mode fiber (SMF) and Multimode fiber (MMF) (2 marks)
4. What do you understand by TCP over IP? (List and explain its working and relevance today) (5 marks)